

Private Multi-Party Inference and an Extensible Research Harness

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01 Why PLLM

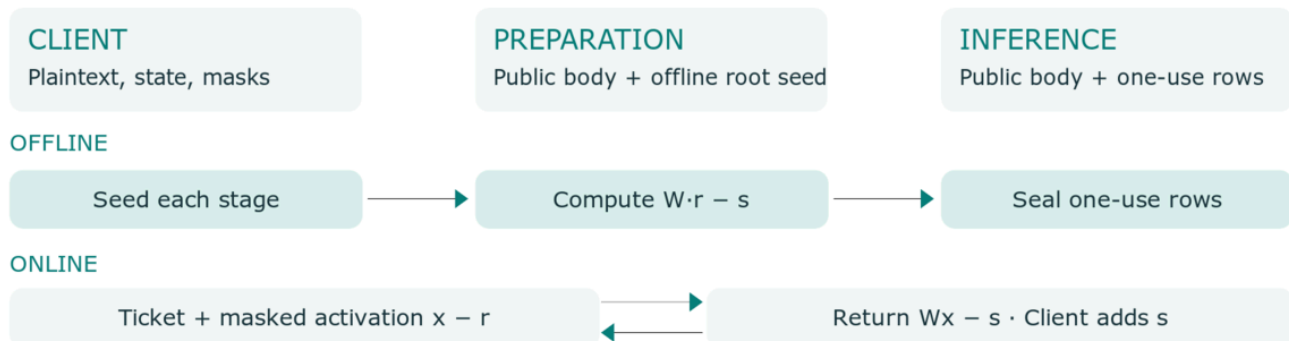
PLLM is a **high-performance private LLM multi-party inference runtime and extensible autonomous research harness**. A hosted endpoint ordinarily receives the request it computes on. PLLM instead aims to give separate operators useful work without granting either remote role the complete activation. Its second job: turn independently implemented research into compatible components that can be tested against a common baseline, rather than accepting paper-reported speedups across different models and trust assumptions.

High performance and economic value are **objectives**, not conclusions drawn from the present measurements. A faster isolated kernel is not automatically a faster, private, deployable model.

02 One private request

Client keeps prompts, tokenization, nonlinear and attention state, masks, sampling, and output. Before generation, trusted Preparation expands fresh stage seeds and uploads one-use masked corrections for public model matrices. Online, Inference receives a ticket and masked activation; Client reconstructs the result. Preparation is idle online. A trusted local gateway can expose Responses and Chat Completions APIs without turning the remote roles into application endpoints.

One response · distinct trust and execution roles



Preparation idle online · both services must follow protocol and not collude

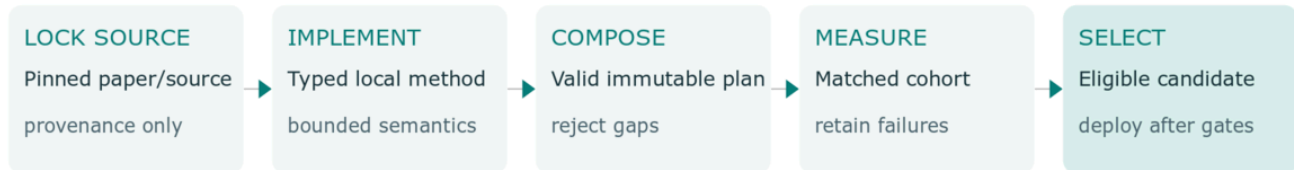
Offline and online work across Client, Preparation, and Inference. Each correction row is used once.

Preparation must follow the protocol, erase masks, and **not collude** with Inference; co-located loopback processes do not demonstrate independent operators. The current public path keeps attention, nonlinear work, and token boundaries local to Client. It does not establish malicious-provider security.

03 Research becomes a candidate capability

Researchers implement pinned papers through typed component or provider contracts. Immutable experiments bind model, composition, workload, and deployment. Bounded search proposes **valid** candidates; matched benchmarks retain failures. Functional, numeric, privacy, quality, and deployment checks gate adoption.

Research-to-capability loop · evidence grows, claims stay scoped



Checks: function · privacy contract · quality · trust · numeric policy
 No automatic paper-to-code translation or measured SOTA winner yet

From paper provenance to a typed implementation, compatible plan, matched evidence, and an eligible deployment choice.

This is an autonomous **experiment-enabling** loop, not automatic paper-to-code translation or security certification. Papers are provenance, never executable dependencies. Static provider manifests support discovery; native provider runtime loading is not yet exposed. New research can improve a deployed choice only when an implementation passes these gates and beats its **eligible matched baseline**. No winner crosses different model, privacy, numeric, hardware, input/output-token, output-cap, or warm-state cohorts by assumption.

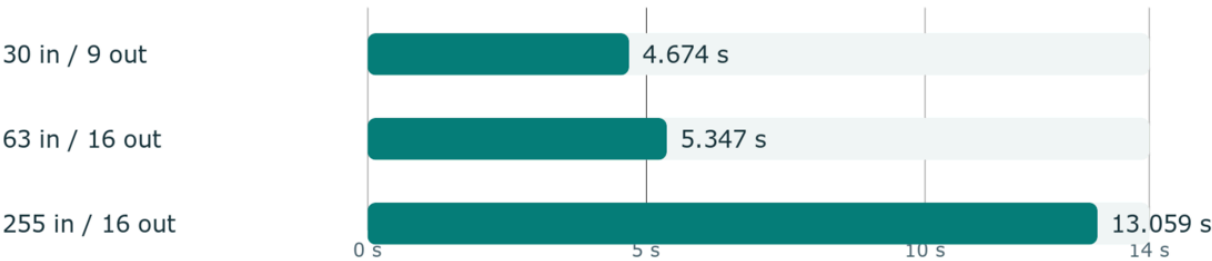
04 Economic value needs an honest denominator

A candidate can be economically useful if it meets the same privacy, model quality, and service constraints while reducing **total cost per successful response**: client work, offline preparation (including burned rows), online compute, network, and operator resources. Latency, throughput, traffic, and quality have different units; a single speed number cannot replace this accounting. PLLM represents metrics and matched benchmark results, but **has not measured prices, energy, or an economic return**, nor established an external state-of-the-art winner. Economic value is a testable hypothesis.

05 Measured baseline; open comparison

One retained study ran Client, Preparation, and Inference on an Apple M5 loopback host using a pinned Qwen2.5-0.5B-Instruct checkpoint. Three warm measurements per **different** input/output-token cohort give these full-response medians (including offline preparation). They are historical measurements from an earlier runtime revision, not current compiler or WAN results.

Historical full-response latency · Qwen2.5-0.5B



Apple M5 loopback · three warm runs/cohort · earlier source revision 277d19f
Different token counts; no competitor, price, energy, WAN or GPU measurement

Historical Qwen2.5-0.5B full-response medians for three distinct token cohorts; no cross-system performance claim.

An optional verification component has only a separate, one-run **tiny-model** functional comparison. Garbling and cache transformations remain bounded or uncomposed, not whole-model Qwen speedups. Next: complete protected decoder coverage, deploy genuinely independent roles, and compare alternatives on the same checkpoint and workload with measured cost and quality. The [technical paper](#) records precise methods, paper-derived components, evidence, and missing comparisons.